

12/3/26

UNIT-8.

analog to Digital and Digital to analog Converters:-

Advantage of Processing digital signal over analog signal.

①. accuracy is More.

②. less Noise

Noise Immunity:

The amount of Noise added to system without affecting its performance (error)

③. More noise immunity.

④. Storage is easy / Retains its originality of signal for long time.

⑤. Reconfigurable.

⑥. use of DSP algorithm is more.

⇒ In DSP it allows plenty of algorithm to process the digital in analog it has

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less algorithms.



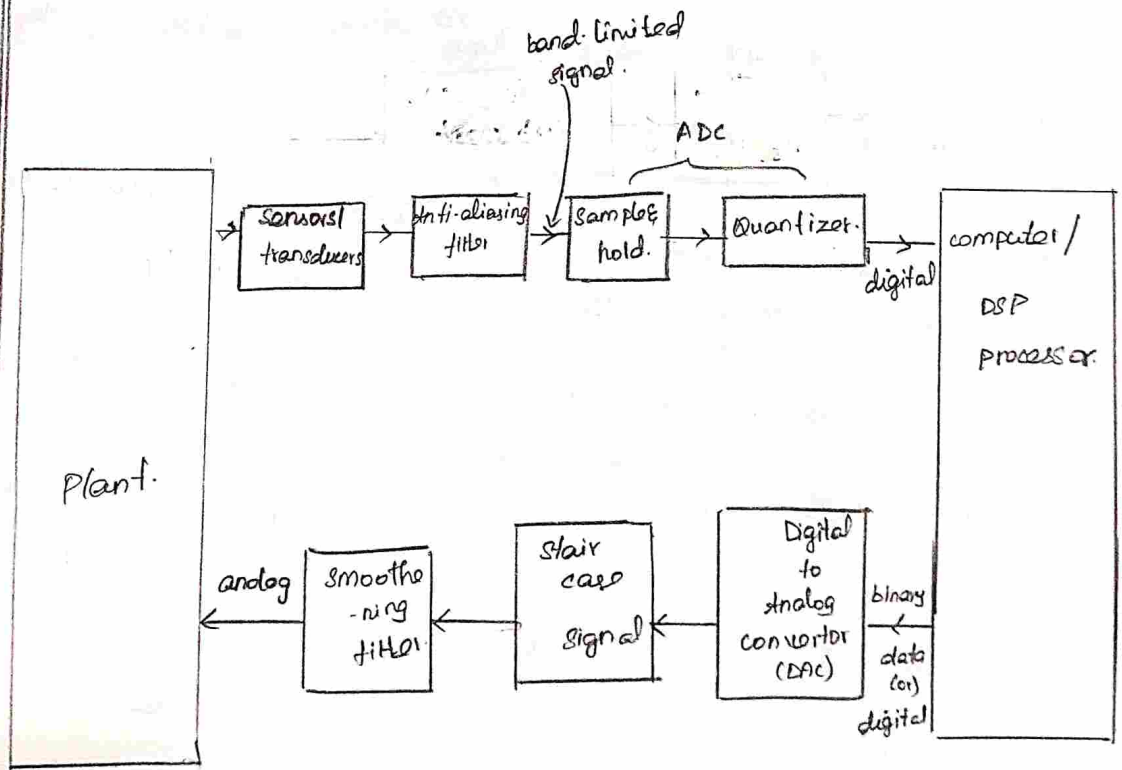
Transducers $\rightarrow$ convert Non-electrical to electrical signal.

* Signal has to comes for only certain limit of frequency - bandlimited freq.

Anti-aliasing filter - it's output should be bandlimited signal.

To convert Analog to digital the signal should be bandlimited then only we go to sampling.

sampling: It is process which can digitalize the analog signal.

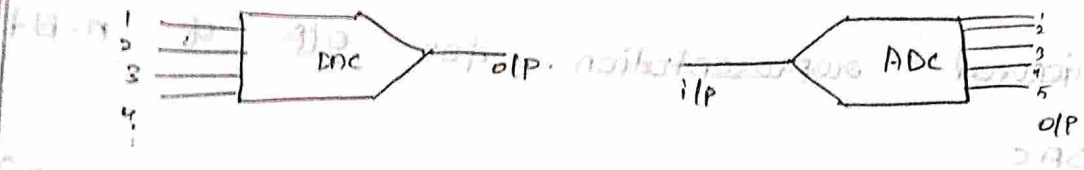


Types:

1. weighted Resistor DAC.
2. R-2R ladder type DAC.
3. Inverted R-2R ladder type DAC.

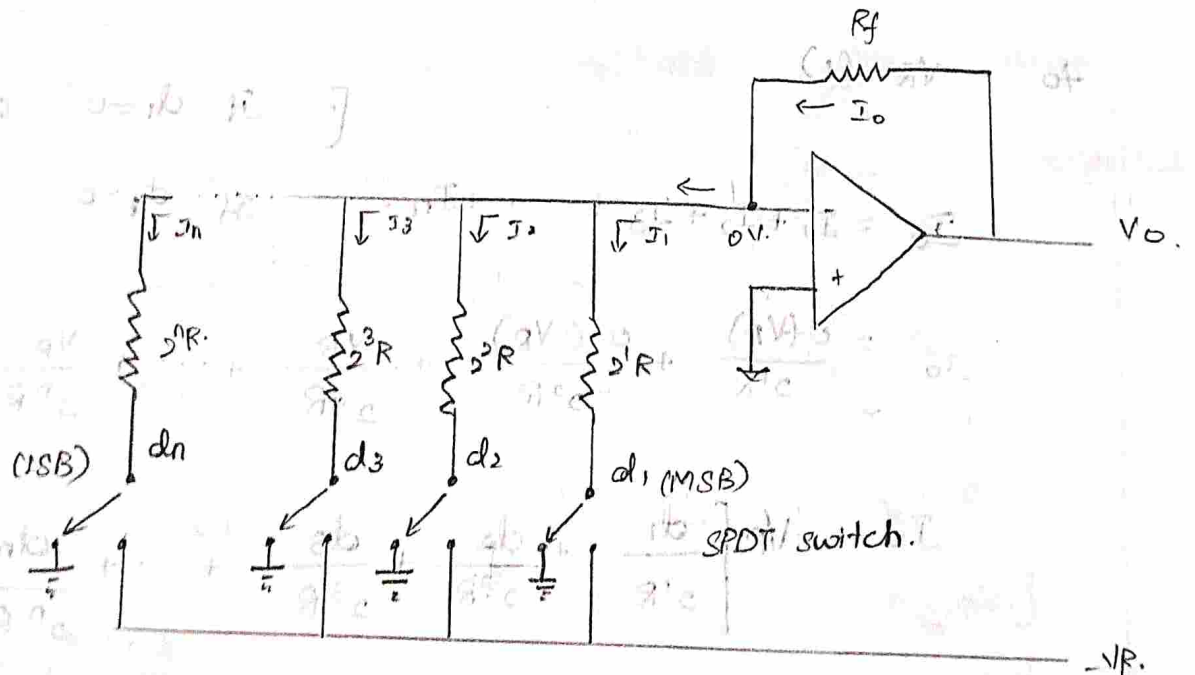
DAC:

we apply 'n' no. of binary i/p gives 1 o/p.

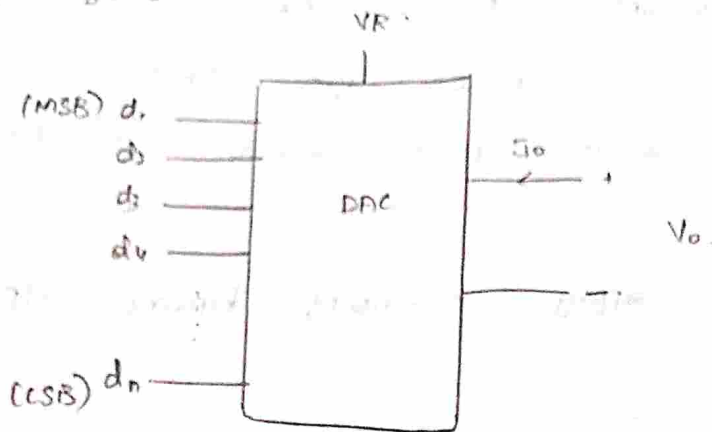


For DAC, configure op-amp as inverting summing amplifier bcoz 'n' no. of i/p's.

①. weighted Resistor DAC.



HOLD use use SFO T - single pos Double through switch.



General representation for O/P of n-bit DAC.

$$V_0 = K V_{FS} [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n 2^{-n}]$$

$V_{FS} \rightarrow$ Full scale voltage.

Assume all Resistor through switch connected to $-V_R$ (1)

[$\therefore$ If $d_1 = 0$ $d_1 \rightarrow$ Ground

$$I_0 = I_1 + I_2 + I_3 + \dots + I_n. \quad \text{If } d_1 = 0 \quad d_1 \rightarrow V_R]$$

$$I_0 = \frac{0 - (-V_R)}{2^1 R} + \frac{0 - (-V_R)}{2^2 R} + \frac{V_R}{2^3 R} + \dots + \frac{V_R}{2^n R}$$

$$I_0 = V_R \left[\frac{d_1}{2^1 R} + \frac{d_2}{2^2 R} + \frac{d_3}{2^3 R} + \dots + \frac{d_n}{2^n R} \right]$$

$$I_0 = \frac{V_R}{R} [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n 2^{-n}]$$

$$V_0 = I_0 R_f$$



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$$V_o = \frac{V_R}{R} [d_1 2^1 + d_2 2^2 + d_3 2^3 + \dots + d_n 2^n] \cdot R_f$$

$$V_o = V_R \cdot \left[\frac{R_f}{R} \right] [d_1 2^1 + d_2 2^2 + \dots + d_n 2^n] \quad \rightarrow \textcircled{2}$$

Compare $\textcircled{1}$ & $\textcircled{2}$.

$$K = \frac{R_f}{R} ; V_{FS} = V_R$$

$K \rightarrow$ Gain,

if $R_f = R$ then Gain, $K = 1$.

for n bit to equivalent analog,

Max value of Resistor $\rightarrow 2^n \cdot R$.

Min Value of Resistor $\rightarrow 2^1 \cdot R$.

Disadvantage/drawback.

* It is not suitable for large range of binary i/p's. [beoz it requires large value of Resistor].

$\rightarrow$ Requires wide Range of R .

* well suited for only $n \leq 8$ bit.

* accuracy is less [due to Resistor].
when ON it provides nearly 0 Resistance, if OFF it provides infinite Resistance. For that reason should be CMOS connected.

realme 12x5G BT connected ON Resistance will not less.



The weightage of bits are.

⑤

$$\begin{array}{lcl} \text{MSB} & \rightarrow & d_1 = \frac{V_{FS}}{2^1} \\ & & \vdots \\ \text{LSB} & \rightarrow & d_n = \frac{V_{FS}}{2^n} \end{array}$$

drawback

- Range of 'R' requires:
1. wide
 2. very careful in selection of switch
 3. Fabrication of Resistance is complex.
 4. accuracy is less.

→ The selection of R is 0.5 kΩ to

avoid overloading effect.

→ Since the current the resistor (is constant) (or) not providing linear output.

→ Since power dissipating involved in weighted type resistor is more (different) leads to non-linear output.

5. Non-linearity output. [due to heat produced in the circuit].

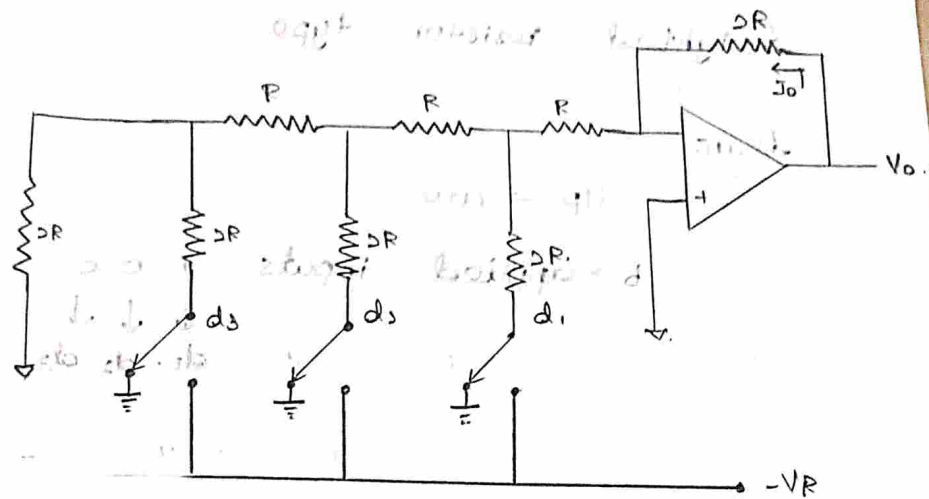
②. R-2R ladder type DAC.

Here the resistors will be either R (or) 2R.



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R-2R ladder type. 3 binary i/p.

assume 3-bits,

$$b = d_1 d_2 d_3.$$

if $R = 2.5 \text{ K}\Omega$.

- * This technique reduce the wide range of Resistor. $\frac{R_C - 0V}{R} = \frac{R_C}{R}$
- * Fabrication is easier compare to weighted resistor type. $\frac{R_C}{R} \gg R_C$
- * Here we require only $R, 2R$ resistors.
- * Current through the resistors are varied by applied inputs.
- * It could not offer constant

voltage

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Power

dissipation

&

heat

produced

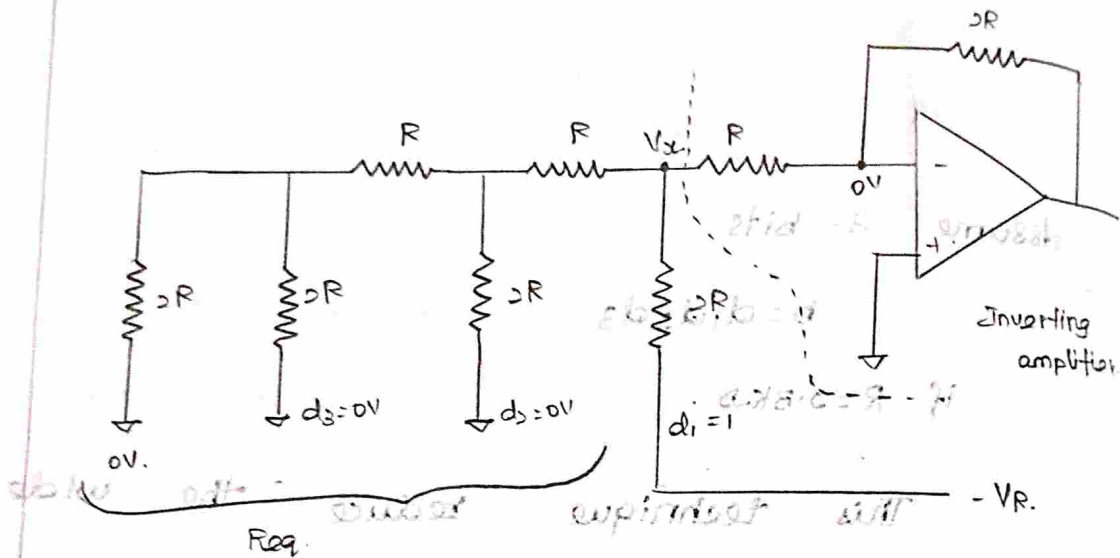
in the circuit is less compared to weighted resistor type.

assume:

$$i_p \rightarrow 100$$

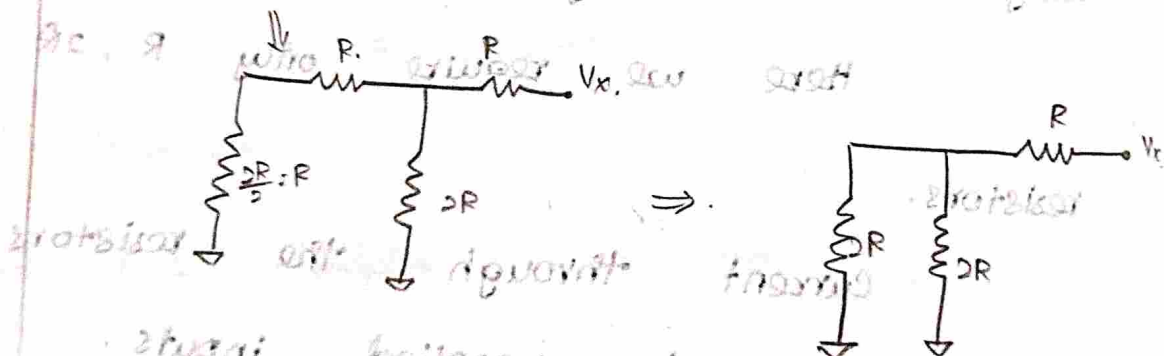
b → applied inputs

$$\begin{matrix} 1 & 0 & 0 \\ \downarrow & \downarrow & \downarrow \\ d_1 & d_2 & d_3 \end{matrix}$$



$$V_0 = -\frac{2R}{R} \cdot V_x \rightarrow \text{①}$$

$$2R \parallel 2R$$



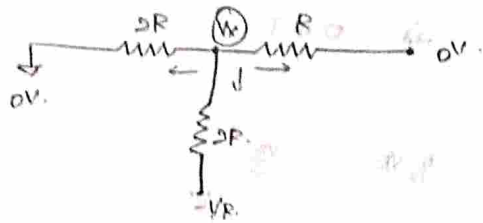
R & R series

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R & R series



Apply Nodal Analysis.

$$\frac{V_x - 0}{2R} + \frac{V_x + V_R}{2R} + \frac{V_x - 0V}{R} = 0.$$

$$V_x \left[\frac{1}{2R} + \frac{1}{2R} + \frac{1}{R} \right] + \frac{V_R}{2R} = 0. \quad \frac{1}{R} + \frac{1}{R} = \frac{2}{R}$$

$$V_x \left[\frac{2}{2R} + \frac{1}{R} \right] + \frac{V_R}{2R} = 0.$$

$$V_x \left[\frac{2}{R} \right] + \frac{V_R}{2R} = 0.$$

$$V_x = -\frac{V_R}{2R} \left[\frac{R}{2} \right]$$

$$\boxed{V_x = -\frac{V_R}{4}} \rightarrow \textcircled{2}$$

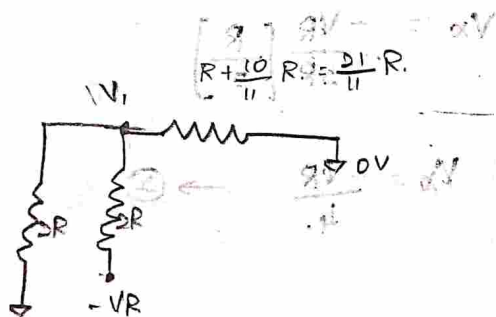
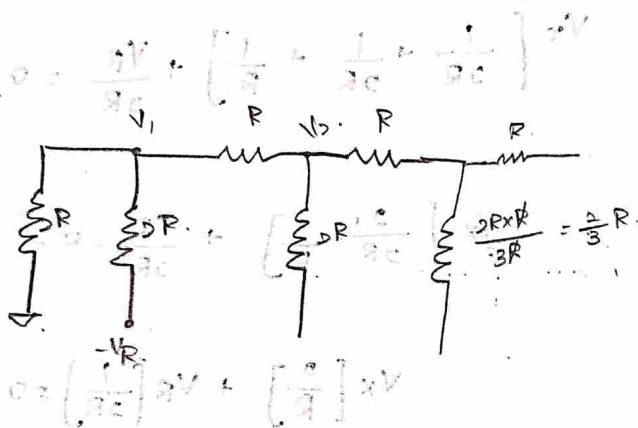
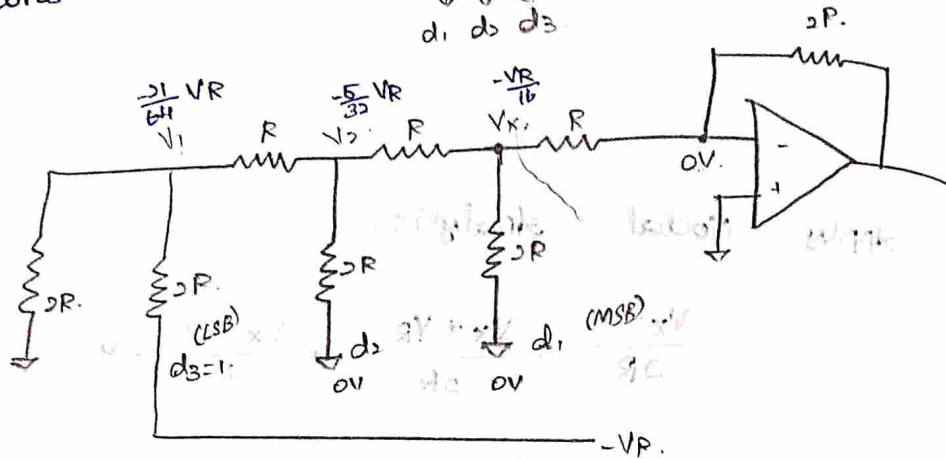
Sub $\textcircled{2}$ in $\textcircled{1}$.

$$V_o = +2 \cdot \left(-\frac{V_R}{4} \right).$$

$$\boxed{V_o = -\frac{V_R}{2}}$$

if $V_R = 5V$ then $V_o = -2.5V$.

consider i/p is 001.



$$\begin{aligned} & \frac{V_1}{2R} + \frac{V_1 + V_R}{2R} + \frac{V_1}{21R} = 0 \\ & \Rightarrow \frac{V_1}{2R} \left[1 + 1 + \frac{1}{21} \right] = -\frac{V_R}{2R} \\ & \Rightarrow \frac{V_1}{2R} \left[\frac{42}{21} + \frac{1}{21} \right] = -\frac{V_R}{2R} \\ & \Rightarrow \frac{V_1}{2R} \left[\frac{43}{21} \right] = -\frac{V_R}{2R} \\ & \Rightarrow V_1 = -\frac{21}{43} V_R \end{aligned}$$

apply Nodal analysis.

$$\frac{V_1}{2R} + \frac{V_1 + V_R}{2R} + \frac{V_1}{21R} = 0$$

$$V_1 \left[\frac{1}{2R} + \frac{1}{2R} + \frac{1}{21R} \right] = -\frac{V_R}{2R}$$

$$V_1 \left[\frac{1}{R} + \frac{1}{21R} \right] = -\frac{V_R}{2R}$$

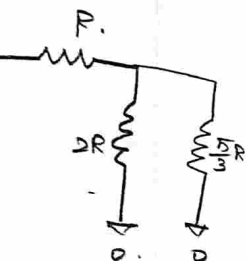


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W.K.T.



W.K.T.

$$\frac{(R)(\frac{5}{3}R)}{(2 + \frac{5}{3})R}$$

$$= \frac{10/3 R}{11/3} = \frac{10}{11} R$$

$$\frac{V_R}{2R}$$

$$\frac{R}{R}$$

$$V_1 \left[1 + \frac{11}{21} \right] = -\frac{V_R}{2}$$

$$V_1 = -\frac{V_R}{2} \left[\frac{21}{82} \right]$$

$$V_1 = -\frac{V_R (21)}{64}$$

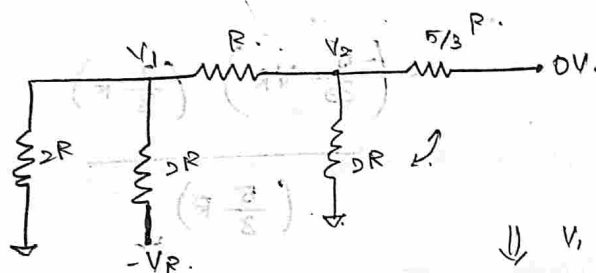
$$V_1 = -\frac{21}{64} V_R$$

W.K.T.

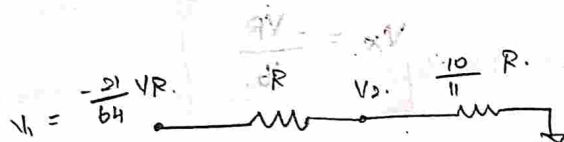
W.K.T.

$$V_0 = -\frac{2R}{R} V_x$$

$$V_0 = -2 V_x \rightarrow \text{①}$$



$$V_1 = -\frac{21}{64} V_R$$



voltage divider rule,

$$V_2 = \frac{\left(-\frac{21}{64} V_R \right)}{\left(R + \frac{10}{11} R \right)} \left(\frac{10}{11} R \right)$$

$$\left(\frac{21}{82} \right) \left(-\frac{21}{64} V_R \right) \left(\frac{10}{11} R \right)$$

$$V_2 = \frac{-\frac{21}{64} V_R \left(\frac{10}{11} \right)}{\left(\frac{21}{11} \right)}$$

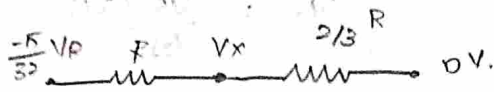
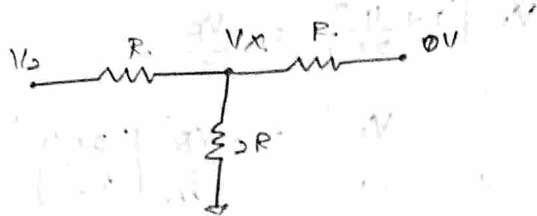
$$V_2 = -\frac{10}{64} V_R$$

$$= -\frac{5}{32} V_R$$

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Voltage divided rule.

$$V_X = \frac{\left(-\frac{5}{32} V_R\right) \left(\frac{2}{3} R\right)}{\left(-R + \frac{2}{3} R\right)}$$

$$= \frac{\left(-\frac{5}{32} V_R\right) \left(\frac{2}{3} R\right)}{\left(-\frac{1}{3} R\right)}$$

$$= \frac{\left(-\frac{5}{32} V_R\right) \left(\frac{2}{3} R\right)}{\left(-\frac{1}{3} R\right)}$$

$$= \frac{5}{16} V_R$$

$$V_X = \frac{5}{16} V_R$$

Sub V_X in equ ①.

$$V_0 = \left(\frac{-2R}{R}\right) V_X$$

$$V_0 = -2 \left(\frac{5}{16} V_R\right)$$

$$V_0 = -\frac{5}{8} V_R$$

if $V_R = 5V$

then

$$V_0 = 0.625 V$$

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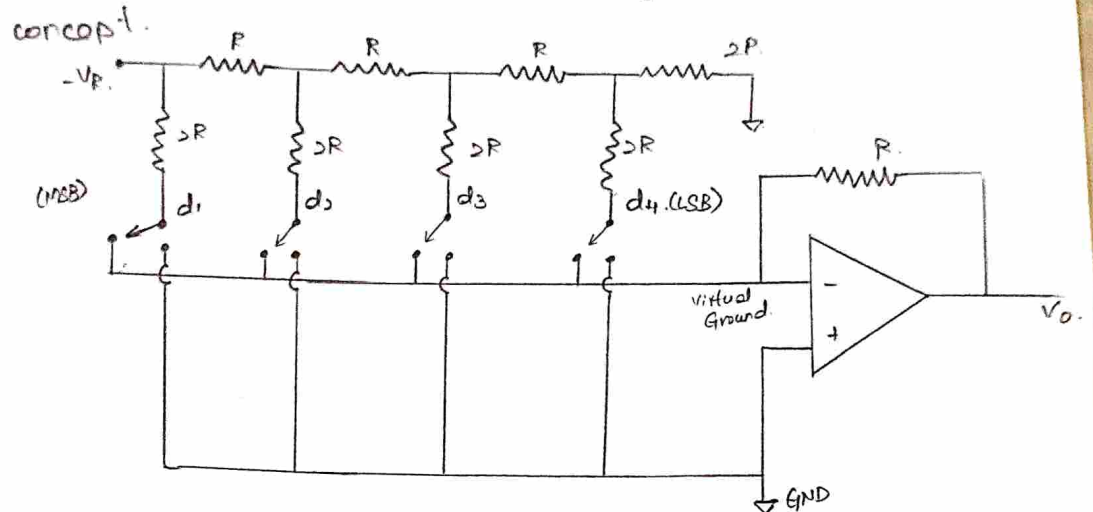
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for b=001



③ Inverted Type R-2R Ladder DAC:

In this case the resistors are connected directly / indirectly to ground for both 0 and 1 using virtual ground concept.



④ Difference b/w R-2R & Inverted R-2R.

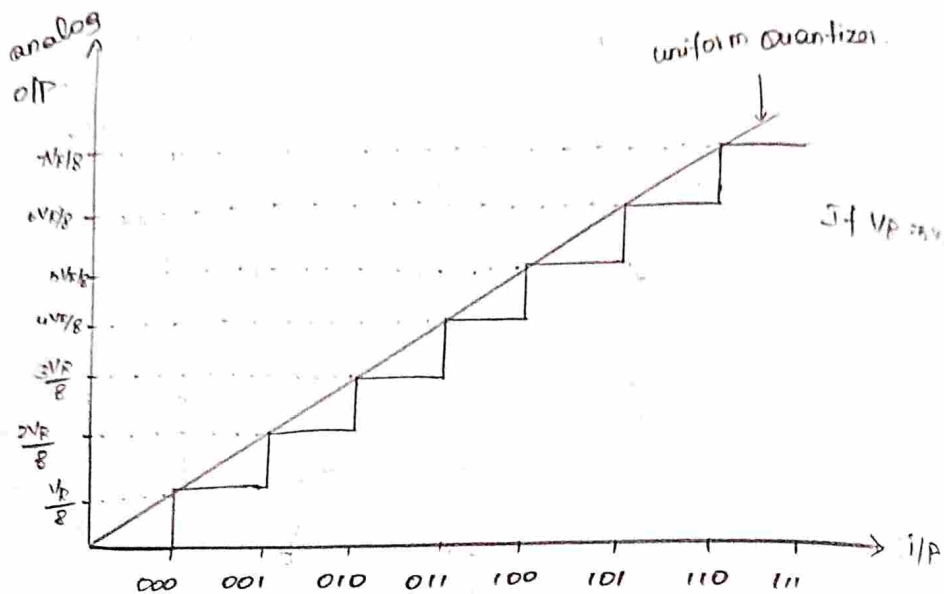
- * With respect to each node it provides equal resistance through the ladder of
- * The current flowing in the each node is constant irrespective of applied i/p.
- * LSB and MSB are exchanged.

It is also called current mode inverted R-2R ladder type DAC.

The fn of circuit is explained by summing the current in each and every node (Resistance).



* o/p of DAC in the form of stair case signal.



Resolution (or)

Resolution (or)

1 LSB (or)

step signal size

$$\left. \begin{array}{l} \text{Resolution (or)} \\ \text{Resolution (or)} \\ \text{1 LSB (or)} \\ \text{step signal size} \end{array} \right\} = \frac{V_R}{2^N} \quad (\text{see follow this})$$

$$R = \frac{V_R}{2^N - 1} \quad [\text{used when } N \text{ is large}]$$

$N \rightarrow$ bit size : 3

$$R: \text{step size} = \frac{V_R}{2^3} = \frac{V_R}{8}$$

For N bit DAC the no. of step

size is $2^N - 1$ level.

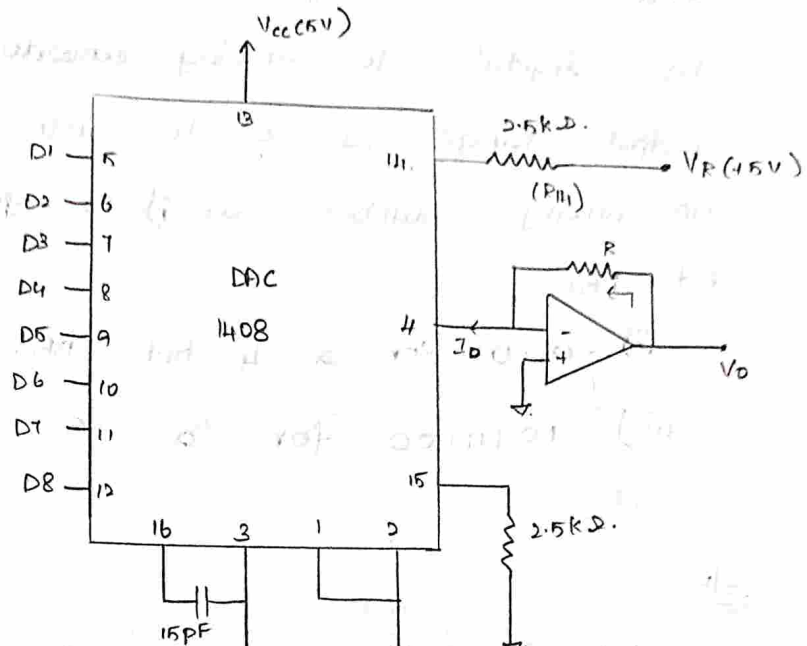
The overall output is

$$V_O = (V_{FS}) - 1 \text{ LSB}$$

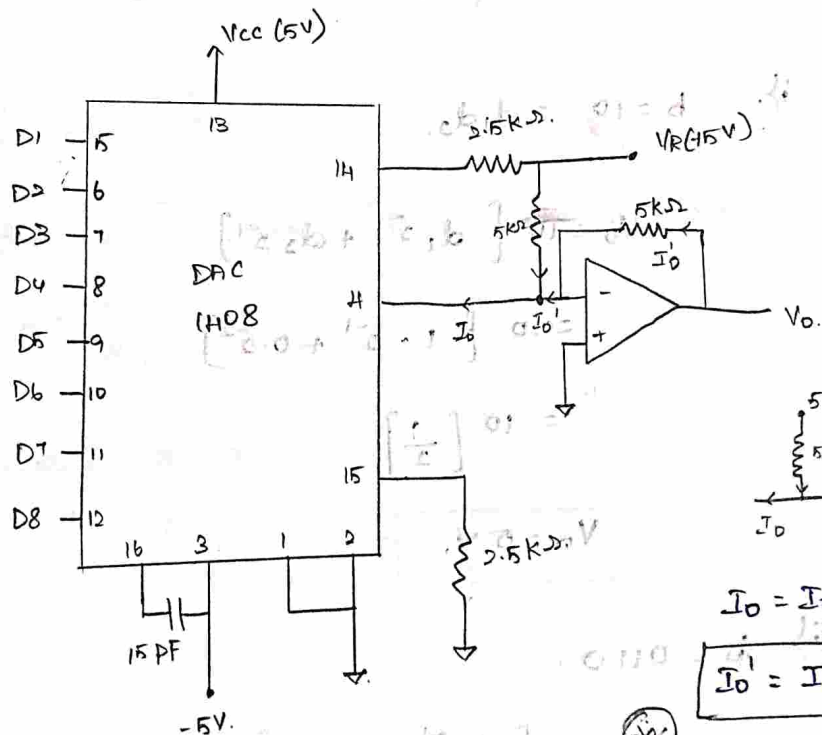


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always.

Pin - Diagram of DAC - 1408 :



$$I_0 = \frac{V_R}{R_{14}} \left[\sum_{i=1}^N d_i 2^{-i} \right] \text{ is obtained from pin 14.}$$



$$I_0 = I_0' + 1 \text{ mA}$$

$$I_0' = I_0 - 1 \text{ mA}$$

$$V_0 = I_0' R_f$$

$$V_0 = I_0 R_f \rightarrow \textcircled{1}$$

$$V_0 = \left[I_0 - \frac{V_R}{5 \text{ k}\Omega} \right] R_f$$

① What output voltage would be produced by digital to analog converter whose output range is 0 to 10V & whose input binary number is i) 10 for a two bit DAC,

ii) 0110 for a 4 bit DAC.

iii) 10111100 for a 8 bit DAC.

In DAC $V_{FS} = V_R$.

Sol:

$$V_o = V_{FS} \left[d_1 \overset{\text{MSB}}{2^{-1}} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n \overset{\text{LSB}}{2^{-n}} \right]$$

$$V_o = V_R \left[d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n 2^{-n} \right]$$

i). $b = 10 = d_1 d_2$.

$$V_o = 10 \left[d_1 2^{-1} + d_2 2^{-2} \right]$$

$$= 10 \left[1 \cdot 2^{-1} + 0 \cdot 2^{-2} \right]$$

$$= 10 \left[\frac{1}{2} \right]$$

$$\boxed{V_o = 5V}$$

ii). $b = 0110$.

$$V_o = 10 \left[d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + d_4 2^{-4} \right]$$

$$= 10 \left[(0) 2^{-1} + (1) 2^{-2} + (1) 2^{-3} + (0) 2^{-4} \right]$$

$$= 10 \left[\frac{1}{4} + \frac{1}{8} \right]$$



$$= 10 \left[\frac{3}{84} \right]$$

$$V_o = \frac{15}{4} \text{ V}$$

$$V_o = 3.75 \text{ V.}$$

iii). $b = 10111100.$

$$V_o = V_R \left[d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + d_4 2^{-4} + d_5 2^{-5} + d_6 2^{-6} + d_7 2^{-7} + d_8 2^{-8} \right].$$

$$V_o = 10 \left[(1) 2^{-1} + (0) 2^{-2} + (1) 2^{-3} + (1) 2^{-4} + (1) 2^{-5} + (1) 2^{-6} + (0) 2^{-7} + (0) 2^{-8} \right]$$

$$= 10 \left[\frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} \right].$$

$$= 10 [0.7343]$$

$$V_o = 7.343 \text{ V}$$

iv). Find its Resolution, LSB, MSB.

$$R = \frac{V_R}{2^N - 1}$$

For binary, $i/p = 2.$

$$R = \frac{10}{2^2 - 1} = \frac{10}{3}$$

$$R = 3.33.$$

For 4 bits.

$$R = \frac{10}{2^4 - 1} = \frac{10}{15} = 0.666.$$



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For 8 bits.

$$R = \frac{10}{2^8 - 1} = \frac{10}{255} = 0.0392$$

$$\begin{aligned} 1 \text{ LSB} &= 2^{-n} V_{FS} \\ 1 \text{ MSB} &= 2^{-1} V_{FS} \end{aligned}$$

For 2 bits,

$$\begin{aligned} 1 \text{ LSB} &= 2^{-2} \times 10 \\ &= \frac{10}{4} \end{aligned}$$

$$\left(\frac{1}{4} = 0.25 V_{FS} \right)$$

$$\begin{aligned} 1 \text{ MSB} &= 2^{-1} \times 10 \\ &= \frac{10}{2} \end{aligned}$$

$$= 5 V.$$

10-bit DAC

For 4 bits,

$$\begin{aligned} 1 \text{ LSB} &= 2^{-4} \times 10 \\ &= \frac{10}{16} \\ &= 0.625 \end{aligned}$$

$$\begin{aligned} 1 \text{ MSB} &= 2^{-1} \times 10 \\ &= 5 \end{aligned}$$

For 8 bits.

$$1 \text{ LSB} = 2^{-8} \times 10$$

$$1 \text{ MSB} = 5$$



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For n bit of weighted Resistor type DAC requires how many resistor.

n resistor required without including R_f .

Analog to Digital converter (ADC).

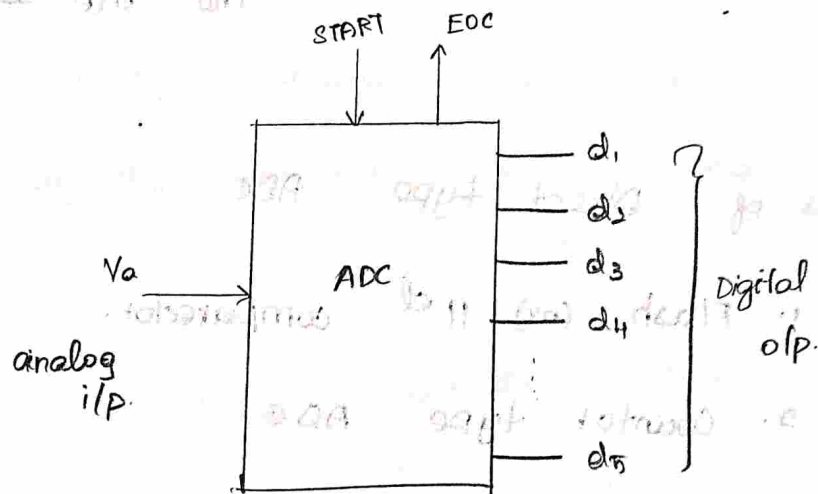
* It uses two control lines

used to identify when the ADC

performs the start and end of conversion.

* START $\rightarrow$ which tells the ADC

when to start conversion process



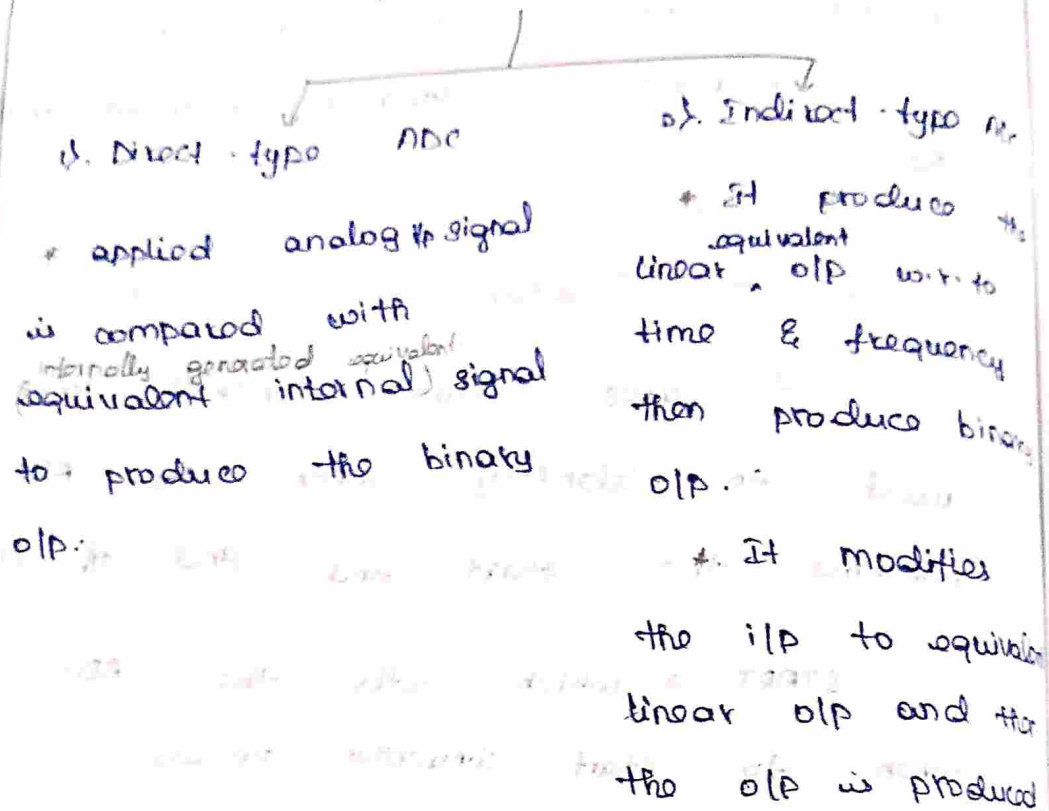
* EOC $\rightarrow$ End of conversion tells when the conversion process of ADC is completed.



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Types of ADC.



Types of Direct-type ADC. (refer ppt slide 2026)

1. Flash (or) ||^l comparator.
 2. Counter type ADC.
 3. Tracking / servo converter.
 4. Successive approximation type ADC.
1. Flash type (or) parallel comparator.

* It is the simplest ADC.

* It is also a fastest ADC.



→ Conversion provided by this is much speed than others).

*. speed is more so conversion time is less. (100ns / 150ns)

→ becoz the conversion mechanism works simultaneously.

*. Simultaneous conversion operation.

It has two circuits.

i) comparators.

ii). Voltage divider circuit.

For n-bit binary (Flash type).

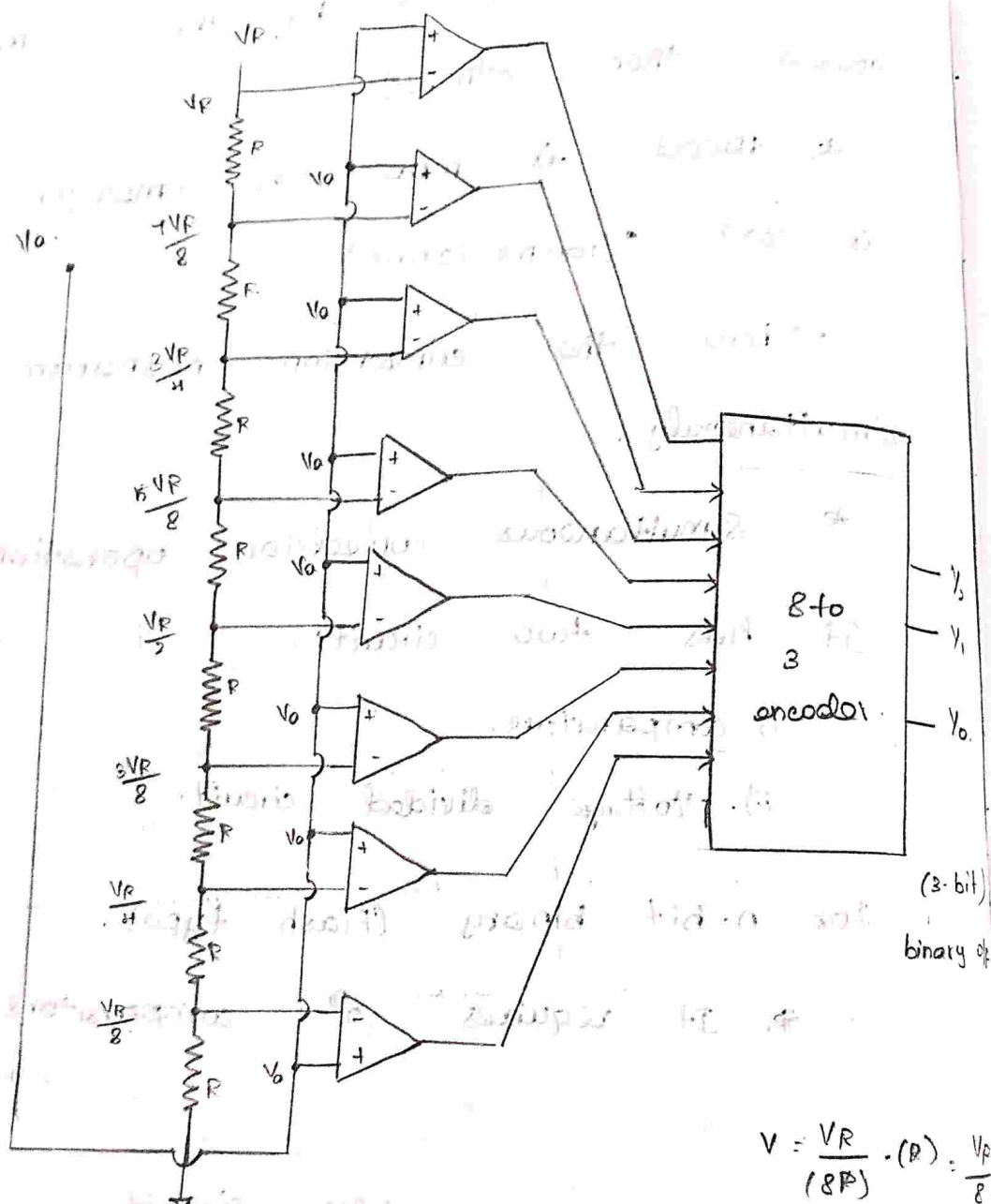
*. It requires $2^n - 1$ comparators.

eg: for 3 bits $2^3 - 1 = 7$ comparators used.

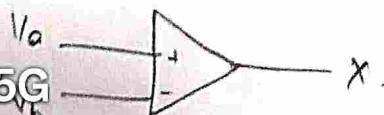
*. Voltage divider circuit.

→ It requires 2^n resistors.

in comparator we compare analog i/p signal & internal equivalent signal.
(+ve terminal)
(-ve terminal)



If $V_a > V_b$, $X = 1$
 $V_a < V_b$, $X = 0$
 $V_a = V_b$, previous state.



applied

0 to

$$\frac{V_R}{8}$$

$$\frac{V_R}{4}$$

$$\frac{3V_R}{8}$$

$$\frac{V_R}{2}$$

$$\frac{5V_R}{8}$$

$$\frac{3V_R}{4}$$

$$\frac{V_R}{8}$$

$$\frac{V_R}{4}$$

$$\frac{3V_R}{8}$$

$$\frac{V_R}{2}$$

$$\frac{5V_R}{8}$$

$$\frac{3V_R}{4}$$

$$\frac{V_R}{8}$$

$$\frac{V_R}{4}$$

$$\frac{3V_R}{8}$$

$$\frac{V_R}{2}$$

$$\frac{5V_R}{8}$$

$$\frac{3V_R}{4}$$

y_2
 y_1
 y_0
 (3-bit)
 binary of

applied analog i/p.	digital o/p											
	x_7	x_6	x_5	x_4	x_3	x_2	x_1	x_0	y_2	y_1	y_0	
0 to $V_R/8$	0	0	0	0	0	0	0	1	0	0	0	
$\frac{V_R}{8}$ to $\frac{V_R}{4}$	0	0	0	0	0	0	1	1	0	0	1	
$\frac{V_R}{4}$ to $\frac{3V_R}{8}$	0	0	0	0	0	1	1	1	0	1	0	
$\frac{3V_R}{8}$ to $\frac{V_R}{2}$	0	0	0	0	1	1	1	1	0	1	1	
$\frac{V_R}{2}$ to $\frac{5V_R}{8}$	0	0	0	1	1	1	1	1	1	0	0	
$\frac{5V_R}{8}$ to $\frac{3V_R}{4}$	0	0	1	1	1	1	1	1	1	0	1	
$\frac{3V_R}{4}$ to $\frac{7V_R}{8}$	0	1	1	1	1	1	1	1	1	1	0	
$\frac{7V_R}{8}$ to V_R	1	1	1	1	1	1	1	1	1	1	1	

* It requires $2^n - 1$ comparator only
 since we using 8:3 encoder so we use
 one additional comparator.

* If question ask how many comparators
 used in Flash-type. answer: $2^n - 1$.

Draw back:

→ This technique is not suited for
 large no. of bits.

→ The requirement of comparator is
 when one bit is increased.



2). Counter - type ADC.

* It has three blocks.

i). Comparator.

→ I/P's * I/P analog signal.

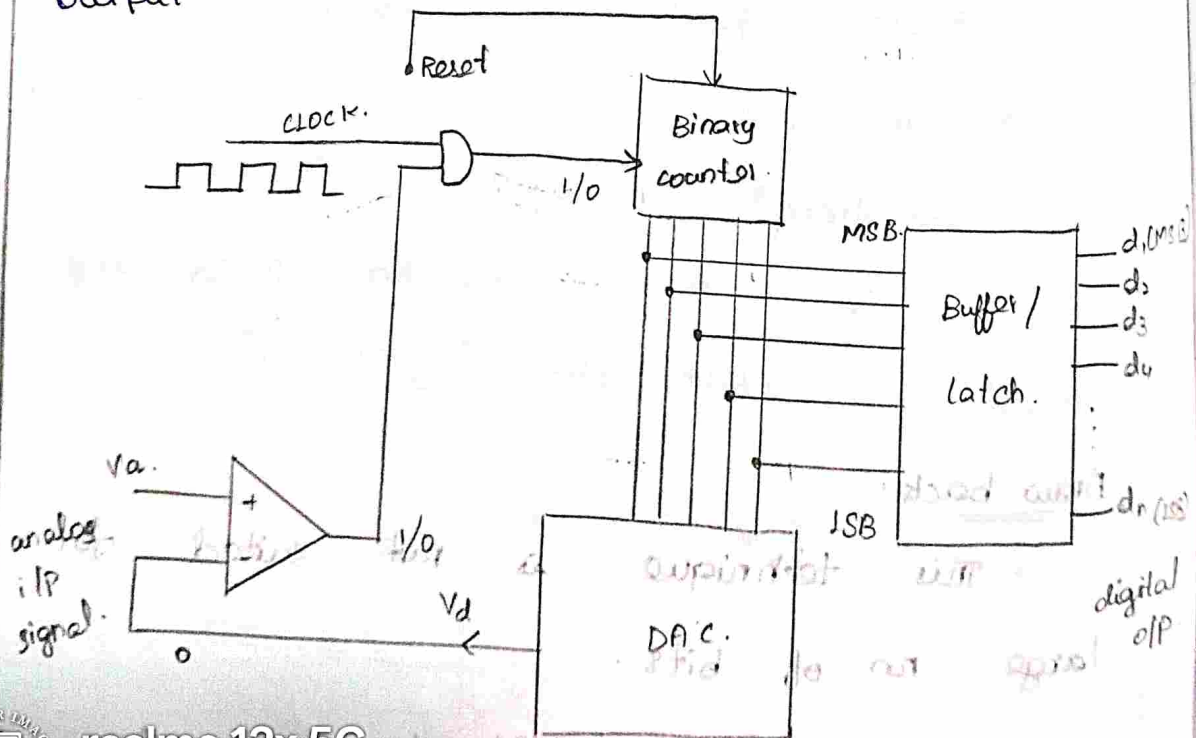
* output of DAC

ii). DAC.

iii). Binary counter. (B.C)

* o/p of Binary counter is connected to DAC.

* Equivalent digital output is produced from Binary counter for the applied analog input and then using Buffer. we get output.



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B.C starts to count in two cases.

we need to release the reset (set 0)



apply analog i/p signal

if $V_a > V_d$, o/p of comparator $\rightarrow 1$.

give clock signal to the AND gate

then the B.C perform counting process

The clock signal can be passed till

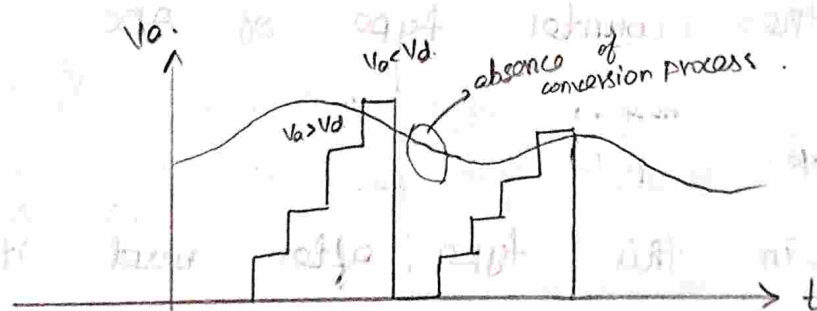
the $V_a > V_d$.

if $V_a < V_d$, o/p of comparator $\rightarrow 0$.

Here clock signal is not passed,

B.C will not count, so it is reset.

after it performs repeatedly.



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* Conversion time is strongly depend on the amplitude of i/p analog signal.

drawback:

1. Conversion time is more.

2. Speed is less.

→ o/p of ADC depends on.

* when counter Reset. (Resetting Processing of counter)

* o/p of DAC.

Conversion time:

$$T_c = (2^N - 1) T_{clk}$$

It is the conversion time offered by the counter type of ADC.

There is chance to absence of conversion

drawback: process b/w after reset & before start of

3. In this type, after reset ^{other clock} it

starts from beginning repeatedly.

3). Servo / tracking counter type ADC:

The binary counter is replaced by up & down counter.

when it counts upward - up counter.
($V_a > V_d$)

when it counts downward - down counter.
($V_d > V_a$).



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ly depend
log signal.

By changing the behaviour of counter
we can reduce the conversion time.

less conversion time compared
to Counter type.

1) Successive approximation Type ADC.

if $V_a > V_d \rightarrow$ the MSB bit is to be
retained and the bit in MSB
is set (1) (o/p of c $\rightarrow 1$)

if $V_a < V_d \rightarrow$ make consent bit as 0 make
next bit as 1. (o/p of comparator $\rightarrow 0$)

11010100 $\rightarrow$ is the digital equivalent
of applied analog i/p signal.

1st give start signal.

V_a	o/p of comparator	V_d (o/p of SAR)
11010100	1	10000000 $V_a > V_d$
	1	11000000 [MSB same $V_a > V_d$ next bit $\rightarrow 1$]
	0	11100000 $V_a < V_d$ MSB $\rightarrow 0$ next $\rightarrow 1$
	1	11010000 $V_a > V_d$
	0	11011000 $V_d > V_a$
	1	11010100

conversion
low start of
clock
it

od by

counter
($V_a > V_d$)
down
($V_d > V_a$)

eg:

Resolution = 4-bit.

$$V_{ref} = 8V.$$

$$V_{in} = 5.2.$$

for 1000,

$$\text{DAC output, } V_o = V_R [d_1 2^1 + \dots + d_n 2^n]$$
$$= 8 \times \frac{1}{2}$$

$$V_o = 4V.$$

along 5.2 V > 4V.

keep.

Next.

for 1100

$$\text{DAC O/P } \Rightarrow V_o = 8 \left[\frac{1}{2} + \frac{1}{4} \right] = 6V.$$

$$V_a < V_d$$

$$5.2 < 6$$

Reset.

then,

10100000111

$$\text{O/P DAC, } V_d = 8 \left[\frac{1}{2} + \frac{1}{8} \right] = 5V.$$

$$5.2 > 5V.$$

keep MSB.

then

1011

$$\text{realme 12x 5G } V_d = 8 \times \left[\frac{1}{2} + \frac{1}{8} + \frac{1}{16} \right] = 5.5V.$$

$$5.2 < 5.5$$

Reset.



then 1010

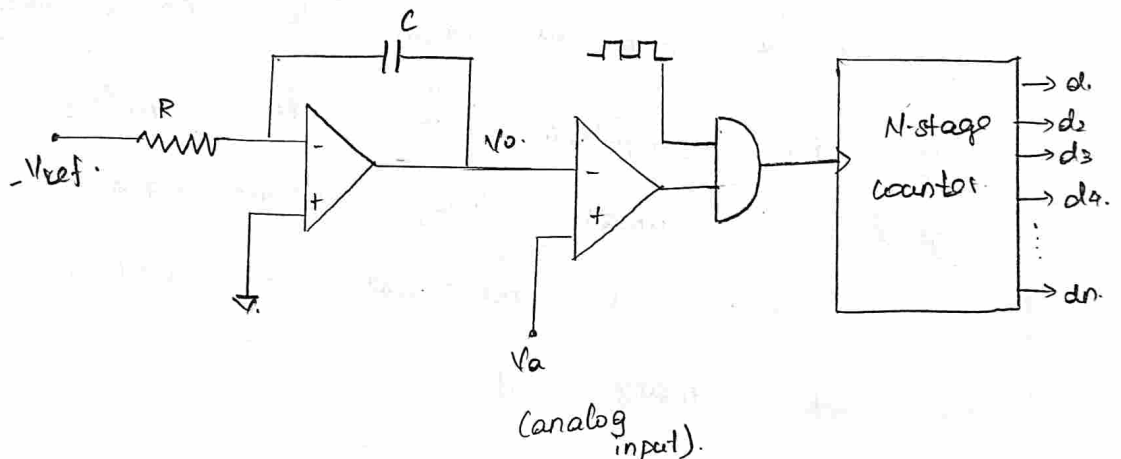
The logic has to be tested for all bits of SAR with the equivalent digital representation.

Indirect type ADC / Integrating type ADC.

types:

1. charge balanced ADC / single slope ADC.
2. Dual slope ADC.

①. charge - balanced ADC:



$$|V_{ref}| > V_a.$$

The magnitude of V_{ref} should be greater than V_a .

$$V_o = -\frac{1}{Rc} \int -V_{ref} \cdot dt.$$

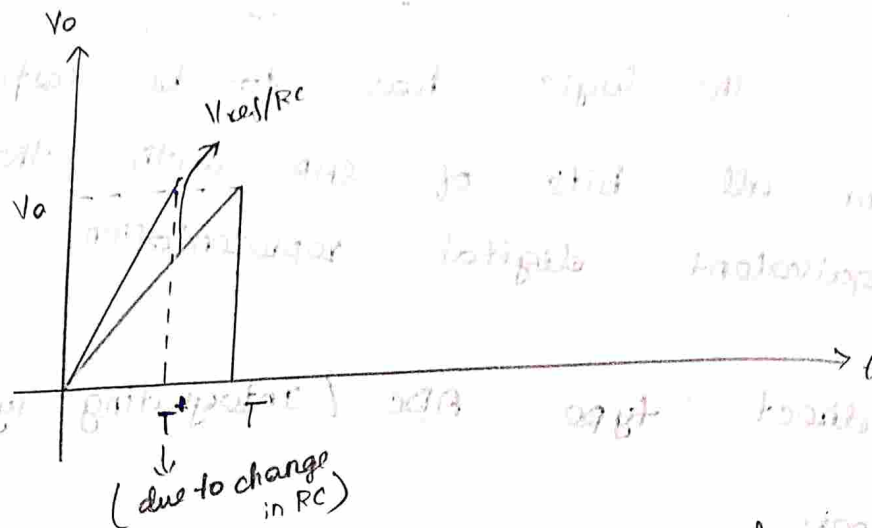
$$V_o = \frac{V_{ref}}{Rc} \cdot T$$

$\rightarrow d_n$



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Output of integrator is connected to comparator circuit.

when $V_o < V_a$, o/p of comparator $\rightarrow 1$.

(o/p of comp is given to AND gate.)

By this converting the i/p analog signal to linear time variant signal

which is connected to counter

we get binary o/p

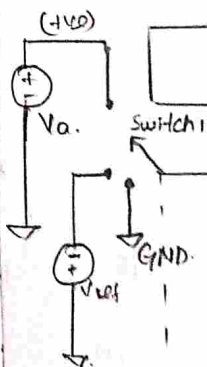
when $V_o > V_a$, o/p of comparator $\rightarrow 0$.

at T , $V_o = 0$.

bcz $V_o > V_a$.

The integrator integrates the

the o/p of comparator



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$V_o < V_a$

The moment at $V_0 > V_a$, o/p of $C \rightarrow 0$.
Counter stops the counting process.

*. If RC is less for the same analog i/p I get t_{HMO} reduced.

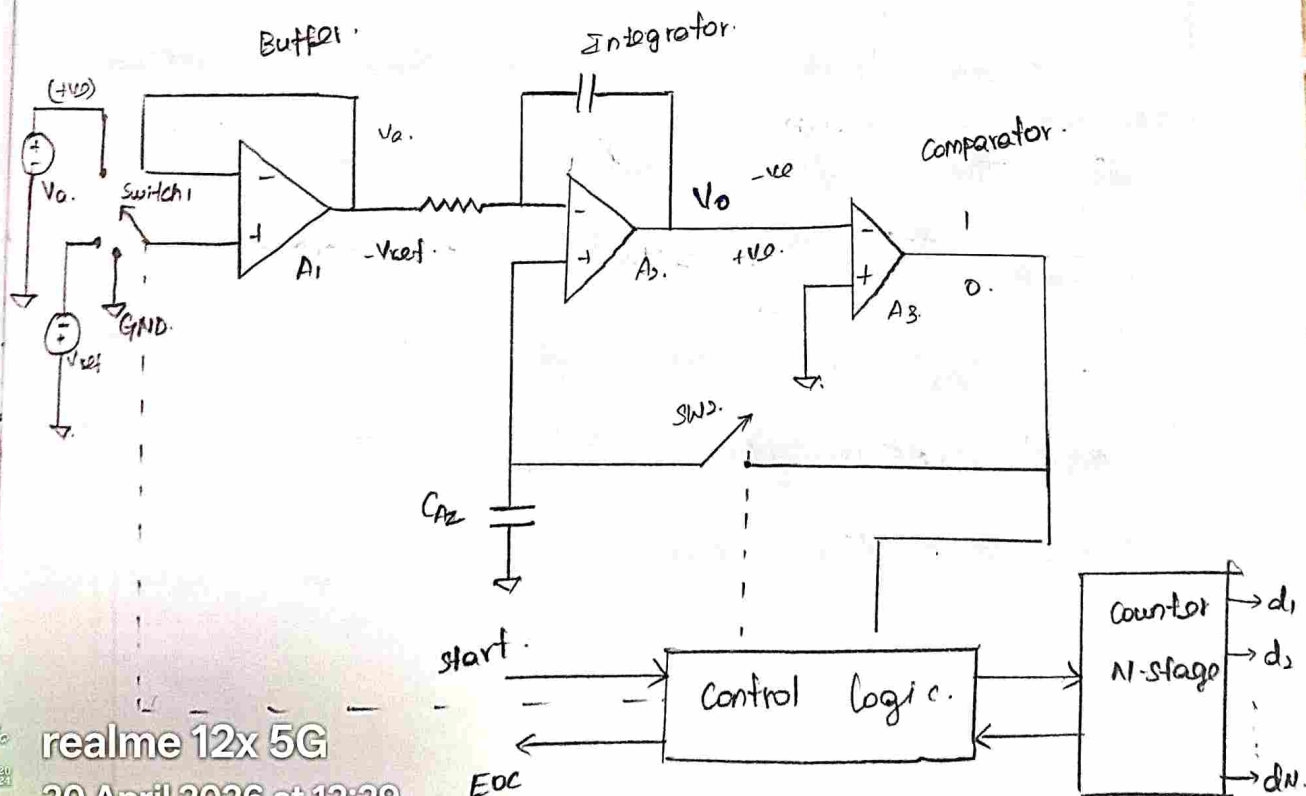
draw back:

*. The same analog i/p, by change in RC we get linear time variant signal is varied.

*. Linearly ~~depen~~ changing ^{time} signal depends on RC for applied analog signal.

* Poor security. ¹ box of.

⑤. Dual-slope ADC. (X)



There are three types of op-amp circuit is used,

1. Buffer (A_1)
2. Integrator (A_2)
3. Comparator (A_3)

⇒ 1st stage - switch 1 is closed, it allows non-inverting terminal allows to ground.

sw 2 is closed

It occurs before the starting process of ADC.

[means we don't apply the i/p to integrator].

→ without applying i/p, there is presence of i/p offset voltage.

→ for that we use C_{A2} . it stores all the offset volt (bcoz o/p of comparator is connected to capacitor)

C_{A2} is used to compensate the effect of i/p offset voltage. appeared in



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before
START

①. $SW_1 \rightarrow \text{END}$, $SW_2 \rightarrow \text{closed}$.

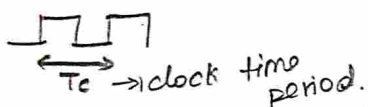
CM $\rightarrow$ Compensate the i/p offset V_{off}
present in A_1, A_2, A_3

②.
2nd stage

$SW_2 \rightarrow \text{open}$, START.

$SW_1 \rightarrow 'V_a'$

O/P of integrator, $V_o = -\frac{1}{RC} \int V_a dt$.



$T_c \rightarrow$ clock time period.

$$V_o = -\frac{V_a}{RC} T_i; T_i = 2^N T_c$$

since V_o is negatively varying slope
signal.

so O/P of comparator is always $\rightarrow 1$.

which allows the counter to

count upto 0 to $2^N - 1$

[during the integration process
the counter will count upto $2^N - 1$].

At end of counting process, the
counter gives the signal to control
logic to allow the SW_1 to connect

Stage-3

⑧. SW1 $\rightarrow -V_{ref}$.

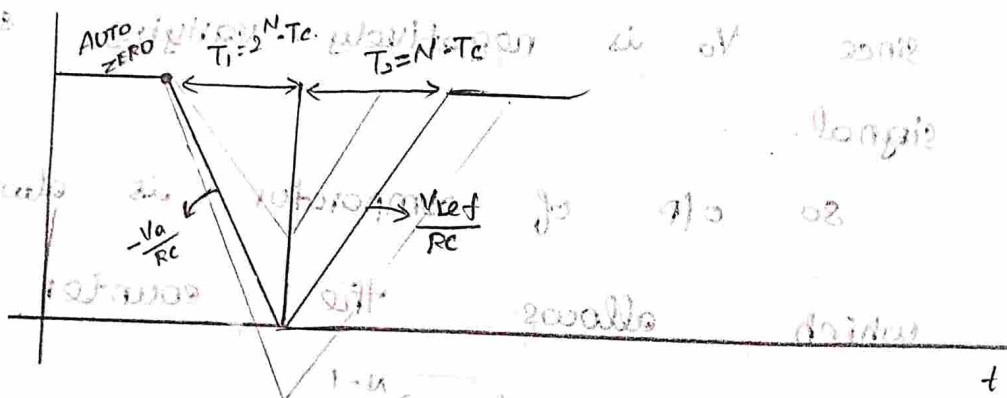
O/P of integrator = $\frac{-1}{RC} \int -V_{ref} dt$.

= $\frac{V_{ref}}{RC} T_s$.

we get +ve Ramp.

The O/P of comparator is 0.
[The moment of crossing 0]

$V_- > V_+$



when O/P of C is 0 the counter

stops counting that time period is

$T_s = N \cdot T_c$

we add the two signal,

linearly changing time w.r.to analog signal



$$\frac{-V_a}{R_C} T_1 + \frac{V_{ref}}{R_C} T_2 = 0$$

$$\frac{V_{ref}}{R_C} T_2 = \frac{V_a}{R_C} T_1$$

$$T_2 = \frac{V_a}{V_{ref}} \cdot T_1$$

→ Time period is depends on V_a, V_{ref} not the R_C .

→ If we change V_a T_2 also changes
 $V_a \uparrow \Rightarrow T_2 \uparrow$

Sub the value of T_1 & T_2 .

$$N \cdot T_C = \frac{V_a}{V_{ref}} \cdot 2^N \cdot T_C$$

$$V_a = \frac{N V_{ref}}{2^N}$$

A dual slope ADC uses a 16-bit counter and a 4 MHz clock rate. The max. i/p voltage is +10 Volt, the max integrator o/p voltage should be -8V when the counter has

through 2^N counts. The capacitor in the integrator is 0.1 μF . Find the



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value of R in the integrator.

sol:

$$V_a = +10V.$$

$$N = 16 \text{ bit}.$$

$$(V_i) V_o = -8V.$$

$$C = 0.1 \mu F.$$

$$\text{Clock rate } f = 4 \text{ MHz}.$$

$$T_c = \frac{1}{4 \times 10^6}.$$

$$T_c = 0.25 \times 10^{-6}.$$

$$T_c = 0.25 \mu\text{sec}.$$

$$-8V = \frac{-V_a}{RC} T_c$$

$$-8 = \frac{-10}{RC} \times 2^N \times T_c.$$

$$8 = \frac{10}{R \times 0.1 \times 10^{-6}} \times 2^{16} \times 0.25 \times 10^{-6}.$$

$$R = \frac{10}{8 \times 0.1} \times 65,536 \times 0.25$$

$$= \frac{10}{0.8} \times 16384.$$

$$= \frac{163840}{0.8}$$

$$R = 204800.$$

$$R = 204.8 \text{ K}\Omega.$$

$$R \approx 205 \text{ K}\Omega.$$



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$$V_a = \frac{N V_{ref}}{2^N} \quad (*)$$

$$V_{ref} = \frac{10 \times 2^{16}}{16.}$$

$$= \frac{10 \times 65536}{16.}$$

$$= \frac{655360}{16.}$$

$$V_{ref} = 40960 \text{ V}$$

if 3-bit,

$$V_a = \frac{10 \times 2^3}{3.}$$

$$V_a = \frac{80}{3} = 26.66.$$

advantage:

→ It could provide best accuracy (high)

drawback:

*. It is only good for slowly varying signal.

*. Conversion time is more.



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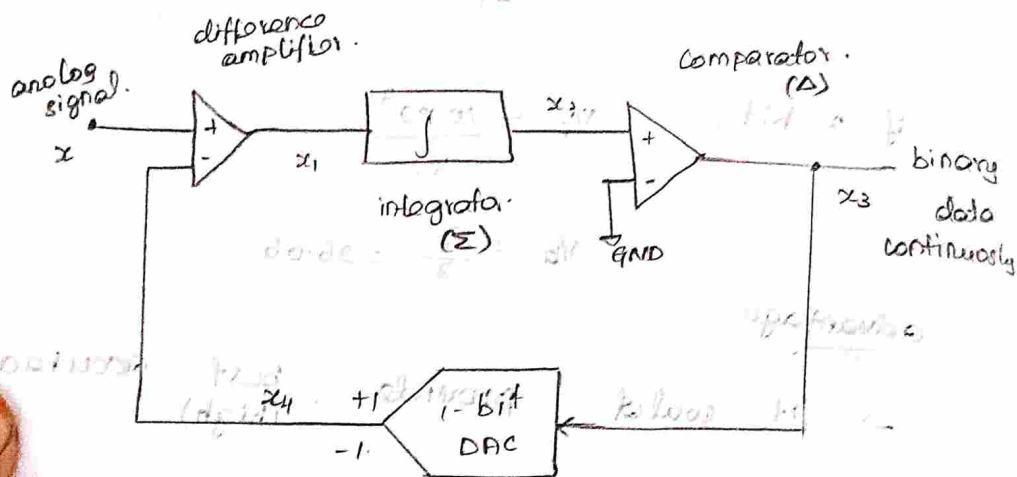
Delta-sigma type ADC: refer ppt



$\Delta \rightarrow$ represents the difference (or) change in parameter in generally.

$\Delta \rightarrow$ represents the comparator.

$\Sigma \rightarrow$ the of sigma is replaced as an integrator circuit.



$\rightarrow$ the o/p of DAC is either +1 (or) -1 designed.

* If it receives 1 as i/p $\rightarrow$ o/p of DAC +1

* If it receives 0 as i/p $\rightarrow$ o/p of DAC -1

Integrator - simply performs addition operation.

The entire function can also be

flowchart:



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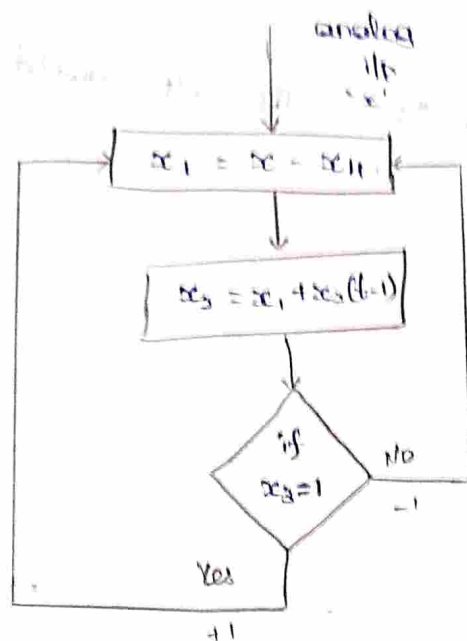
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ADC | DAC

1. Resolution
2. Linearity
3. Accuracy
4. Settling
5. Stability

①. Resolution

changes



x_1 → applied analog sig.
 $x_2(l-1)$ → previous output.
 x_{11} → o/p of DAC
 x_2 → o/p of integrator

If $x_1 \rightarrow +\infty$, o/p → group of 1's.
 If $x_1 \rightarrow -\infty$, o/p → group of 0's.

ADC / DAC specifications:

1. Resolution
2. Linearity.
3. accuracy.
4. settling time.
5. stability.

①. Resolution.

It tells how the output changes when 1 LSB change in i/p.



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